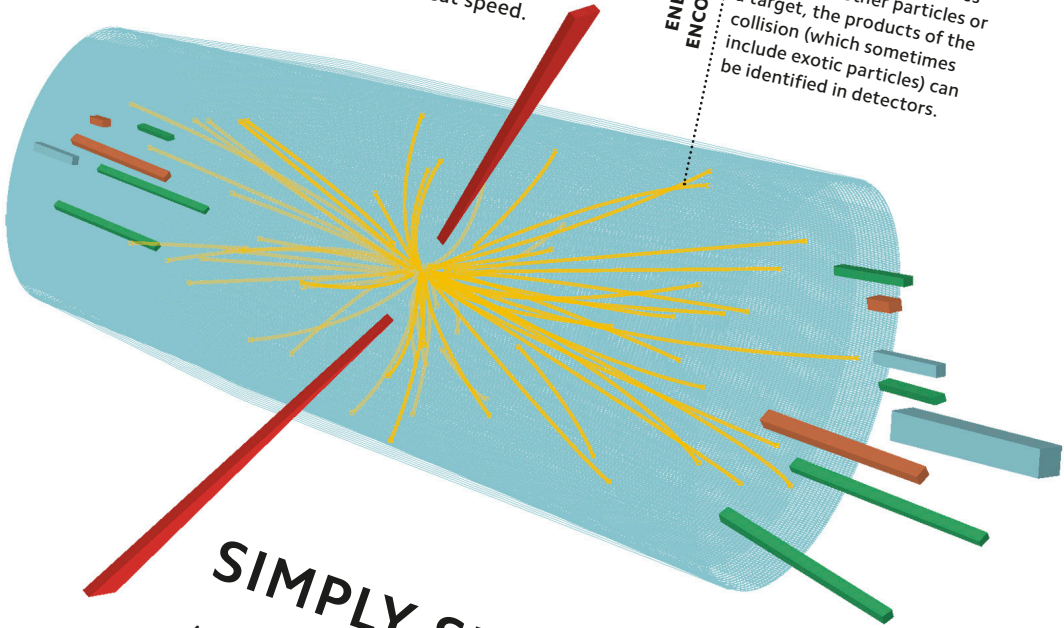


Colliding particles

At particle accelerators, such as the Large Hadron Collider at CERN, charged particles are smashed into each other at great speed.

ENERGETIC ENCOUNTER

When high-energy particles collide with other particles or a target, the products of the collision (which sometimes include exotic particles) can be identified in detectors.



SIMPLY SMASHING

A particle accelerator propels particles to extreme speeds and energies, allowing physicists to probe the smallest objects in nature. Modern particle accelerators use changing electromagnetic fields to accelerate and guide charged particles to almost light speed, and smash them into targets. Smashing particles together causes them to break apart, fuse, and—at extreme energies—create the exotic matter that briefly existed in the instant after the Big Bang.